1-var Single-variable tools

L'Hôpital for $0/0$ or ∞/∞: $\lim f/g = \lim f'/g'$; **re-check the form** before every use. $0\cdot\infty$: rewrite as $f/(1/g)$. $1^\infty, 0^0, \infty^0$: take ln first; $\lim_{x\to\infty}(1+a/x)^{bx} = e^{ab}$.

Maclaurin (often faster near 0) $e^x = 1 + x + x^2/2 + x^3/6$ ▪ $\sin x = x - x^3/6$ ▪ $\cos x = 1 - x^2/2 + x^4/24$ ▪ $\ln(1+x) = x - x^2/2 + x^3/3$ ▪ $\tan x = x + x^3/3$ ▪ $\arctan x = x - x^3/3$ ▪ $(1+x)^k = 1 + kx + k(k-1)x^2/2$.

FTC with moving limits for $a = a(x)$, $b = b(x)$:

$$\frac{d}{dx}\int_a^b g(t)\,dt = g(b)b' - g(a)a'$$

Integration substitution (**change the limits**); parts $\int u\,dv = uv - \int v\,du$, choosing u by LIATE; $\int \ln x\,dx = x\ln x - x$.

Improper integrals replace the bad endpoint by a limit; split at an interior singularity. $\int_1^\infty x^{-p}\,dx$ converges $\Leftrightarrow p > 1$; $\int_0^1 x^{-p}\,dx \Leftrightarrow p < 1$. Ready values: $\int_0^\infty x^n e^{-ax}\,dx = n!/a^{n+1}$ ▪ $\int_0^1 x^n \ln x\,dx = -1/(n+1)^2$ ▪ $\int_0^1 x^n(\ln x)^2\,dx = 2/(n+1)^3$ ▪ $x^a(\ln x)^k \to 0$ as $x \to 0^+$ for $a > 0$.

Applications area $\int(\text{top} - \text{bottom})\,dx$ ▪ volume of revolution: discs $\pi\int R^2$, washers $\pi\int(R^2 - r^2)$, shells $2\pi\int(\text{radius})(\text{height})$ ▪ arc length $\int\sqrt{1+y'^2}\,dx$.

Implicit curves $F(x,y) = 0 \Rightarrow dy/dx = -F_x/F_y$. Horizontal tangent: $F_x = 0$ with $F = 0$, **and** $F_y \neq 0$. Related rates: differentiate in t first, substitute after.

11 Vectors, lines, planes

Dot product $\mathbf{u}\cdot\mathbf{v} = \sum u_i v_i = |\mathbf{u}|\,|\mathbf{v}|\cos\theta$; $\mathbf{u}\perp\mathbf{v} \Leftrightarrow \mathbf{u}\cdot\mathbf{v} = 0$. Projection $\text{proj}_\mathbf{v}\,\mathbf{u} = ((\mathbf{u}\cdot\mathbf{v})/|\mathbf{v}|^2)\mathbf{v}$, and $\mathbf{u} - \text{proj}_\mathbf{v}\,\mathbf{u} \perp \mathbf{v}$.

Cross product $\mathbf{u}\times\mathbf{v} = \langle u_2v_3 - u_3v_2, u_3v_1 - u_1v_3, u_1v_2 - u_2v_1\rangle$, perpendicular to both. $|\mathbf{u}\times\mathbf{v}| = |\mathbf{u}|\,|\mathbf{v}|\sin\theta$ is the parallelogram area (triangle: half). Box volume $|(\mathbf{u}\times\mathbf{v})\cdot\mathbf{w}|$.

Line through P_0 along $\mathbf{v}$: $\mathbf{r} = P_0 + t\mathbf{v}$; through P, Q take $\mathbf{v} = Q - P$. **Plane** with normal $\mathbf{n} = \langle a,b,c\rangle$ through P_0: $ax + by + cz = \mathbf{n}\cdot P_0$.

Normal of a plane through three points: $\overrightarrow{PQ}\times\overrightarrow{PR}$ ▪ containing a line $(P_0,\mathbf{v})$ and a point A: $\mathbf{v}\times\overrightarrow{P_0A}$ ▪ containing A, B and perpendicular to a plane with normal $\mathbf{m}$: $\overrightarrow{AB}\times\mathbf{m}$. Line where two planes meet: direction $\mathbf{n}_1\times\mathbf{n}_2$; for a point on it set $z = 0$ and solve.

Line meets plane substitute $x(t), y(t), z(t)$ into the plane, solve for t, back-substitute. Line $\parallel$ plane $\Leftrightarrow \mathbf{v}\cdot\mathbf{n} = 0$; line $\perp$ plane $\Leftrightarrow \mathbf{v}\parallel\mathbf{n}$; planes parallel $\Leftrightarrow$ normals parallel; angle between planes = angle between normals.

Distances point S to the plane $ax + by + cz = d$: $|ax_S + by_S + cz_S - d| / |\mathbf{n}|$ ▪ S to the line $(P_0,\mathbf{v})$: $|\overrightarrow{P_0S}\times\mathbf{v}| / |\mathbf{v}|$ ▪ parallel planes: $|d_1 - d_2| / |\mathbf{n}|$ ▪ skew lines: $|\overrightarrow{P_1P_2}\cdot(\mathbf{v}_1\times\mathbf{v}_2)| / |\mathbf{v}_1\times\mathbf{v}_2|$.

Curve $\mathbf{r}(t)$ velocity $\mathbf{r}'(t)$, speed $|\mathbf{r}'(t)|$. Tangent line at t_0: $\mathbf{r}(t_0) + s\mathbf{r}'(t_0)$. Position from velocity: $\mathbf{r}(t) = \mathbf{r}(t_0) + \int_{t_0}^t \mathbf{v}$. Arc length $\int_a^b |\mathbf{r}'(t)|\,dt$: simplify $|\mathbf{r}'|^2$ before taking the root. Collide: same t; paths cross: separate parameters t, s.

13.3 Partials and the chain rule

Partials f_x: differentiate in x, every other variable held constant. Clairaut: $f_{xy} = f_{yx}$, so $f_x = P$, $f_y = Q$ is solvable only if $P_y = Q_x$.

Rebuild f from f_x, f_y ❶ $f = \int f_x\,dx + g(y)$. ❷ Differentiate in y and match f_y: this gives $g'(y)$. ❸ Integrate g', then fix the constant from the given value.

Chain rule sum over every path. If $z = z(u,v,w)$ and each depends on (x,y): $z_y = z_u u_y + z_v v_y + z_w w_y$. An unknown $v(x,y)$ is fine: only v_x, v_y are needed. Substitute u, v, w back at the end.

Along a path $(d/dt)f(\mathbf{r}(t)) = \nabla f(\mathbf{r}(t))\cdot\mathbf{r}'(t)$: **evaluate ∇f at the point $\mathbf{r}(t_0)$.**

Implicit $F(x,y,z) = 0 \Rightarrow z_x = -F_x/F_z$, $z_y = -F_y/F_z$.

Limits in two variables continuous $\Rightarrow$ substitute. No limit: two paths ($y = mx$, $y = kx^2$) give different values. Limit 0: in polar the bound $\to 0$ as $r \to 0$ for every θ.

13.5 Gradient and tangent planes

Directional derivative $D_\mathbf{u}f = \nabla f\cdot\mathbf{u}$ with $\mathbf{u}$ a **unit** vector: divide the direction by its length first. Largest rate $|\nabla f|$ along ∇f; smallest $-|\nabla f|$ along $-\nabla f$; zero perpendicular to ∇f.

Tangent plane to $F(x,y,z) = c$ at P_0: $\nabla F(P_0)\cdot(\mathbf{x} - P_0) = 0$; normal line $P_0 + t\nabla F(P_0)$. Graph $z = f(x,y)$: $z = f(a,b) + f_x(x-a) + f_y(y-b)$, which is also the linearization; $df = f_x\,dx + f_y\,dy$.

Tangent plane parallel to a given plane (normal $\mathbf{n}$) ❶ Write $\nabla F = \lambda\mathbf{n}$; a constant component fixes λ. ❷ Solve for the coordinates it determines. ❸ Use $F = c$ for the one that remains.

Also tangent to the curve $F = c$, $G = k$: direction $\nabla F\times\nabla G$ ▪ ∇f is perpendicular to the level set through the point ▪ gradient descent: $\mathbf{x}_{k+1} = \mathbf{x}_k - h\nabla f(\mathbf{x}_k)$, $h > 0$.

13.7 Critical points and extrema

Critical points solve $f_x = 0 = f_y$: factor and split into cases ($6xy = 0 \Rightarrow x = 0$ or $y = 0$); **never divide by a variable**.

Second derivative test $D = f_{xx}f_{yy} - f_{xy}^2$ at the point; only the second partials there are needed.

$D > 0$, $f_{xx} < 0$	$D > 0$, $f_{xx} > 0$	$D < 0$	$D = 0$
local max	local min	saddle	no verdict

Largest or smallest value on a closed bounded region ❶ Interior critical points. ❷ Boundary: Lagrange, or parametrize each piece. ❸ Corners. Compare f at **all** the candidates.

Least squares fit $y = ag(x) + bh(x)$: solve $a\sum g_i^2 + b\sum g_ih_i = \sum g_iy_i$ and $a\sum g_ih_i + b\sum h_i^2 = \sum h_iy_i$. Line $y = mx + b$: $m\sum x_i^2 + b\sum x_i = \sum x_iy_i$ and $m\sum x_i + nb = \sum y_i$.

Taylor to order 2 at (a,b), with $h = x - a$, $k = y - b$: $f \approx f + f_xh + f_yk + (f_{xx}h^2 + 2f_{xy}hk + f_{yy}k^2)/2$.

13.8 Lagrange multipliers

Extremize f on $g = c$ solve $\nabla f = \lambda\nabla g$ together with $g = c$. Written out: $f_x = \lambda g_x$, $f_y = \lambda g_y$ (and $f_z = \lambda g_z$), $g = c$. Then evaluate f at **every** solution: the largest value is the maximum, the smallest the minimum. Two constraints: $\nabla f = \lambda\nabla g + \mu\nabla h$.

Solving the system factor each equation and split into cases (a variable is 0, or λ takes a value) ▪ eliminate λ: $f_xg_y = f_yg_x$ ▪ symmetric in x, y: try $x = y$ ▪ $f = xyz$: multiply the three equations by x, y, z.

Shortcuts max of $ax + by + cz$ on the sphere of radius R: $R\sqrt{a^2+b^2+c^2}$ ▪ max of $ax + by$ on $px^2 + qy^2 = k$: $\sqrt{k(a^2/p + b^2/q)}$ ▪ point of the plane $\mathbf{n}\cdot\mathbf{x} = d$ nearest the origin: $(d/|\mathbf{n}|^2)\mathbf{n}$ ▪ with $x + y + z$ fixed, $x^ay^bz^c$ is largest when $x : y : z = a : b : c$.

∫ Integrals and derivatives

Inner integrals $\int re^{-r^2}\,dr = -e^{-r^2}/2$ ▪ $\int r/(1+r^2)\,dr = \ln(1+r^2)/2$ ▪ $\int r\sqrt{a^2-r^2}\,dr = -(a^2-r^2)^{3/2}/3$ ▪ $\int\cos^n\varphi\sin\varphi\,d\varphi = -\cos^{n+1}\varphi/(n+1)$ ▪ $\int\sin^3\varphi\,d\varphi = -\cos\varphi + (\cos^3\varphi)/3$ ▪ $\int\sec^2 = \tan$ ▪ $\int\tan = \ln|\sec|$ ▪ $\int\sec = \ln|\sec + \tan|$.

Standard forms $\int dx/(a^2+x^2) = (1/a)\arctan(x/a)$ ▪ $\int dx/\sqrt{a^2-x^2} = \arcsin(x/a)$ ▪ $\int dx/\sqrt{x^2 \pm a^2} = \ln|x + \sqrt{x^2 \pm a^2}|$ ▪ $\int\sqrt{a^2-x^2}\,dx = (x/2)\sqrt{a^2-x^2} + (a^2/2)\arcsin(x/a)$.

Derivatives $(\tan x)' = \sec^2 x$ ▪ $(\sec x)' = \sec x\tan x$ ▪ $(\arcsin x)' = 1/\sqrt{1-x^2}$ ▪ $(\arctan x)' = 1/(1+x^2)$ ▪ $(a^x)' = a^x\ln a$.

14 Which method?

Rectangle, product $g(x)h(y)$ split into single integrals.

Triangle, region between curves sketch; take the order that needs one integral.

Inner integral impossible reverse the order.

Disc, annulus, x^2+y^2 polar. **Symmetric region, odd integrand** 0.

Parallelogram, curves $g=c_1, c_2$ substitute $u=g$, $v=h$.

Cylinder, paraboloid, x^2+y^2 **in space** cylindrical, z innermost.

Ball, cone, $x^2+y^2+z^2$ spherical.

14.1 Double integrals

Iterated integral $\iint_R f\,dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f\,dy\,dx$ (vertical strips) $= \int_c^d \int_{h_1(y)}^{h_2(y)} f\,dx\,dy$ (horizontal strips). **Outer limits are constants**; inner limits depend only on the outer variable. Rectangle with $f=g(x)h(y)$: $(\int g\,dx)(\int h\,dy)$.

Setting up ❶ Sketch the region; equate the curves to find where they meet. ❷ Push an arrow through the region in the inner direction: lower limit where it enters, upper limit where it leaves. ❸ If the entry or exit curve changes along the region, take the other order (or split).

Reverse the order when the inner integral has no elementary antiderivative (e^{x^2}, $\sin y/y$, $1/\sqrt{y^4+1}$). ❶ Read the region off the limits: $a\le x\le b$, $g_1(x)\le y\le g_2(x)$. ❷ Sketch it and invert the boundary curves ($y=x^{1/3} \Leftrightarrow x=y^3$). ❸ Describe it the other way; the new inner integral supplies the factor a substitution needs.

Tools scaling: $\int_0^a f(kx)\,dx = (1/k)\int_0^{ka} f(u)\,du$ ▪ symmetry: region symmetric in x and integrand odd in $x \Rightarrow 0$; even $\Rightarrow$ twice the half ▪ line through (x_1,y_1), (x_2,y_2) as x in terms of y: $x = x_1 + \frac{x_2-x_1}{y_2-y_1}(y-y_1)$.

Uses area $\iint 1\,dA$ ▪ volume $\iint(z_{\text{top}}-z_{\text{bot}})\,dA$ ▪ average $(1/\text{area})\iint f\,dA$ ▪ centroid $\bar{x}=(1/A)\iint x\,dA$, so $\iint x\,dA=\bar{x}A$ (triangle: centroid = mean of the vertices) ▪ mass $M=\iint\delta\,dA$, $\bar{x}=(1/M)\iint x\delta\,dA$.

14.4 Polar coordinates

$x=r\cos\theta$, $y=r\sin\theta$, $x^2+y^2=r^2$, $dA=r\,dr\,d\theta$. Use for discs, annuli, sectors, and integrands in x^2+y^2.

Curve	In polar
disc of radius a	$0\le r\le a,\ 0\le\theta\le 2\pi$
line $y=mx$	$\theta=\arctan m$ ($y=x$: $\pi/4$; $y=\sqrt{3}x$: $\pi/3$)
$x^2+y^2=2ax$	$r=2a\cos\theta,\ -\pi/2\le\theta\le\pi/2$
$x^2+y^2=2ay$	$r=2a\sin\theta,\ 0\le\theta\le\pi$
$x=a$; $y=b$	$r=a\sec\theta$; $r=b\csc\theta$

$y=\sqrt{2x-x^2}$ is the upper half of $r=2\cos\theta$: $0\le\theta\le\pi/2$. Area inside $r(\theta)$: $\int_\alpha^\beta (r^2/2)\,d\theta$. Gaussian: $\int_0^\infty e^{-ax^2}\,dx=\sqrt{\pi/a}/2$.

Trig $\sin^2\theta=(1-\cos 2\theta)/2$ ▪ $\cos^2\theta=(1+\cos 2\theta)/2$ ▪ $\sin 2\theta=2\sin\theta\cos\theta$ ▪ $1+\tan^2\theta=\sec^2\theta$.

Powers of sin, cos $\int_0^{\pi/2}\sin^n=\int_0^{\pi/2}\cos^n$ equals, for $n=2,\dots,6$: $\pi/4$, $2/3$, $3\pi/16$, $8/15$, $5\pi/32$. Over $[0,\pi]$ double the sin values. $\int_0^{2\pi}\cos^2=\int_0^{2\pi}\sin^2=\pi$; $\int_0^{2\pi}\sin\cos=0$.

14.8 Change of variables

$$\iint_R f\,dx\,dy=\iint_G f\,|J|\,du\,dv,\qquad J=\frac{\partial(x,y)}{\partial(u,v)}=x_uy_v-x_vy_u$$

❶ Choose u,v so that the boundary curves become $u=$ const and $v=$ const.

❷ Compute the easy Jacobian $\partial(u,v)/\partial(x,y)=u_xv_y-u_yv_x$; then J is its reciprocal.

❸ Write $f\,|J|$ in u,v; the awkward factor often cancels.

❹ Limits from the boundary curves, in the order asked for.

Common choices $u=x+y$, $v=x-y$: $|J|=1/2$ ▪ $u=xy$, $v=y/x$: $\partial(u,v)/\partial(x,y)=2v$, with $x^2=u/v$, $y^2=uv$ ▪ $u=x^2-y^2$, $v=xy$: $2(x^2+y^2)$ ▪ ellipse $x=ar\cos\theta$, $y=br\sin\theta$: $dA=abr\,dr\,d\theta$ ▪ three variables: $dV=|\partial(x,y,z)/\partial(u,v,w)|\,du\,dv\,dw$.

14.5 Triple integrals

Limits outer limits are constants; middle limits depend on the outer variable only; inner limits may depend on both. With z innermost: $z_{\text{bot}}(x,y)\le z\le z_{\text{top}}(x,y)$, then a double integral over the shadow in the xy-plane.

Shortcuts box with a product integrand: product of three single integrals ▪ region symmetric under $x\to -x$ and integrand odd in $x \Rightarrow 0$ ▪ ball centred at O: $\iiint x^2=\iiint y^2=\iiint z^2=(1/3)\iiint\rho^2$ ▪ slicing: $\iiint g(z)\,dV=\int g(z)A(z)\,dz$, with $A(z)$ the cross-section area.

Uses volume $\iiint 1\,dV$ ▪ mass $M=\iiint\delta\,dV$ ▪ centre of mass $\bar{x}=(1/M)\iiint x\delta\,dV$ ▪ average $(1/V)\iiint f\,dV$.

14.7 Cylindrical coordinates

$x=r\cos\theta$, $y=r\sin\theta$, $z=z$, $dV=r\,dz\,dr\,d\theta$. Use when the shadow is a disc or annulus, or the integrand contains x^2+y^2.

Surfaces cylinder $x^2+y^2=a^2$: $r=a$ ▪ cone $z=\sqrt{x^2+y^2}$: $z=r$ ▪ paraboloid $z=x^2+y^2$: $z=r^2$ ▪ sphere: $z=\pm\sqrt{a^2-r^2}$ ▪ plane $z=6+x$: $z=6+r\cos\theta$.

Order z innermost, from the bottom surface to the top one; then r and θ over the shadow (equate the two surfaces for its radius). $\int_0^{2\pi}\cos\theta\,d\theta=\int_0^{2\pi}\sin\theta\,d\theta=0$ removes the slanted part of a plane.

14.7 Spherical coordinates

$x=\rho\sin\varphi\cos\theta$, $y=\rho\sin\varphi\sin\theta$, $z=\rho\cos\varphi$, $r=\rho\sin\varphi$, $x^2+y^2+z^2=\rho^2$, $dV=\rho^2\sin\varphi\,d\rho\,d\varphi\,d\theta$. Here $\rho\ge 0$; $\varphi\in[0,\pi]$ is measured **down from the $+z$-axis**; θ is as in polar.

Surface	In spherical
sphere $x^2+y^2+z^2=a^2$	$\rho=a$
sphere $x^2+y^2+z^2=2az$	$\rho=2a\cos\varphi$, $\varphi\le\pi/2$
cone $z=k\sqrt{x^2+y^2}$	$\tan\varphi=1/k$
plane $z=c$	$\rho=c\sec\varphi$
cylinder $x^2+y^2=a^2$	$\rho=a\csc\varphi$
xy-plane; above it	$\varphi=\pi/2$; $\varphi\le\pi/2$

Cone angles: $k=1$ gives $\varphi=\pi/4$; $\sqrt{3}$ gives $\pi/6$; $1/\sqrt{3}$ gives $\pi/3$. Above the cone means $0\le\varphi\le$ that angle.

Ranges whole ball: $0\le\rho\le a$, $0\le\varphi\le\pi$, $0\le\theta\le 2\pi$; first octant: $\varphi,\theta\in[0,\pi/2]$. Radial integrand: $\iiint_{\rho\le a} g(\rho)\,dV=4\pi\int_0^a g(\rho)\rho^2\,d\rho$. $\int_0^\alpha\sin\varphi\,d\varphi=1-\cos\alpha$.

Volumes ball $(4/3)\pi a^3$ ▪ cone $(1/3)\pi r^2h$ ▪ cylinder πr^2h ▪ inside $\rho=a$ and above the cone $\varphi=\alpha$: $(2/3)\pi a^3(1-\cos\alpha)$ ▪ ellipsoid $(4/3)\pi abc$ ▪ tetrahedron with intercepts a,b,c: $abc/6$.

! Before you box the answer

Jacobian present? r in polar and cylindrical, $\rho^2\sin\varphi$ in spherical, $|J|$ in a substitution (absolute value, rewritten in u,v).

Limits outer limits are numbers; after reversing, read both limits again from the sketch. $r=2a\cos\theta$ runs over $-\pi/2\le\theta\le\pi/2$, not $[0,2\pi]$.

Do not mix up $\sqrt{x^2+y^2}=r=\rho\sin\varphi$ and $\sqrt{x^2+y^2+z^2}=\rho$; in cylindrical $x^2+y^2+z^2=r^2+z^2$. Saddle $\Leftrightarrow D<0$, whatever the sign of f_{xx}.

Rewrite, do not compute give the limits, the integrand and the Jacobian, in the stated order of integration.

Form fraction in lowest terms ▪ point as (a,b,c) ▪ $a\sqrt{b}$ with b as small as possible ▪ in terms of π ▪ integer powers of e. Answer what is asked: "saddle points" is not "maximum value".